Code:

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// Program to find the inverse of a matrix using Gauss-Jordan Elimination

#include<iostream>
#include<iomanip>
#include<cmath>
using namespace std;

int main()
{
   cout<<fixed<<setprecision(2);

   int i,j,k,n;
   double mat[100][200], t;

   cout<<"Enter the dimension of the matrix: ";
   cin>>n;

   cout<<"Enter the elements of the matrix:\n";
   for(i=0;i<n;i++)
      for(j=0;j<n;j++)
         cin>>mat[i][j];

   // Form augmented matrix [A | I]
   for(i=0;i<n;i++)
   {
      for(j=n;j<2*n;j++)
      {
         if(j==i+n)
            mat[i][j]=1;
         else
            mat[i][j]=0;
      }
   }

   cout<<"\nAugmented Matrix:\n";
   for(i=0;i<n;i++)
   {
```

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   for(j=0;j<2*n;j++)
      cout<<setw(8)<<mat[i][j];
   cout<<endl;
}

// Partial Pivoting
for(i=0;i<n;i++)
{
   for(k=i+1;k<n;k++)
   {
      if(fabs(mat[k][i])>fabs(mat[i][i]))
      {
         for(j=0;j<2*n;j++)
         {
            double temp=mat[i][j];
            mat[i][j]=mat[k][j];
            mat[k][j]=temp;
         }
      }
   }
}

// Gauss-Jordan Elimination
for(i=0;i<n;i++)
{
   if(mat[i][i]==0)
   {
      cout<<"\nThe matrix is not invertible.\n";
      return 0;
   }

   for(j=0;j<n;j++)
   {
      if(j!=i)
      {
         t=mat[j][i]/mat[i][i];

         for(k=0;k<2*n;k++)
```

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                mat[j][k]-=t*mat[i][k];
            }
        }
    }

    // Make pivot elements equal to 1
    for(i=0;i<n;i++)
    {
        t=mat[i][i];

        for(j=0;j<2*n;j++)
            mat[i][j]/=t;
    }

    cout<<"\nInverse Matrix:\n";

    for(i=0;i<n;i++)
    {
        for(j=n;j<2*n;j++)
            cout<<setw(8)<<mat[i][j];
        cout<<endl;
    }

    return 0;
}
```

Output:

```
Enter the dimension of the matrix: 2
Enter the elements of the matrix:
1 2
3 4

Augmented Matrix:
   1.00    2.00    1.00    0.00
   3.00    4.00    0.00    1.00

Inverse Matrix:
  -2.00    1.00
```

```
   1.50   -0.50

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Process exited after 5.315 seconds with return value 0
Press any key to continue . . .
```